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Track: Data Engineering

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Task 1

1. What problem is the organization trying to solve?

The business had a problem because 5 of their systems were independent and doesn't connect to each other. They want a one centralized platform for ease of use and reliability for reporting.

2. Who will eventually use the data?

The management, and operations team in the business like sales, finance, customer service, etc.

3. What are the five operational data sources?

Customer System, Product Catalog, Order Management System, Payment Gateway, Delivery System.

4. What information does each source contain?

Customers = profile/account info

Products = catalog details

Orders = what was bought and its status

Payments = what was actually paid

Deliveries = shipment/fulfillment status.

5. Which fields appear to connect one system to another?

customer_id links Customers to Orders; order_id links Orders to both Payments and Deliveries; Products aren't directly linked yet (no line-items table).

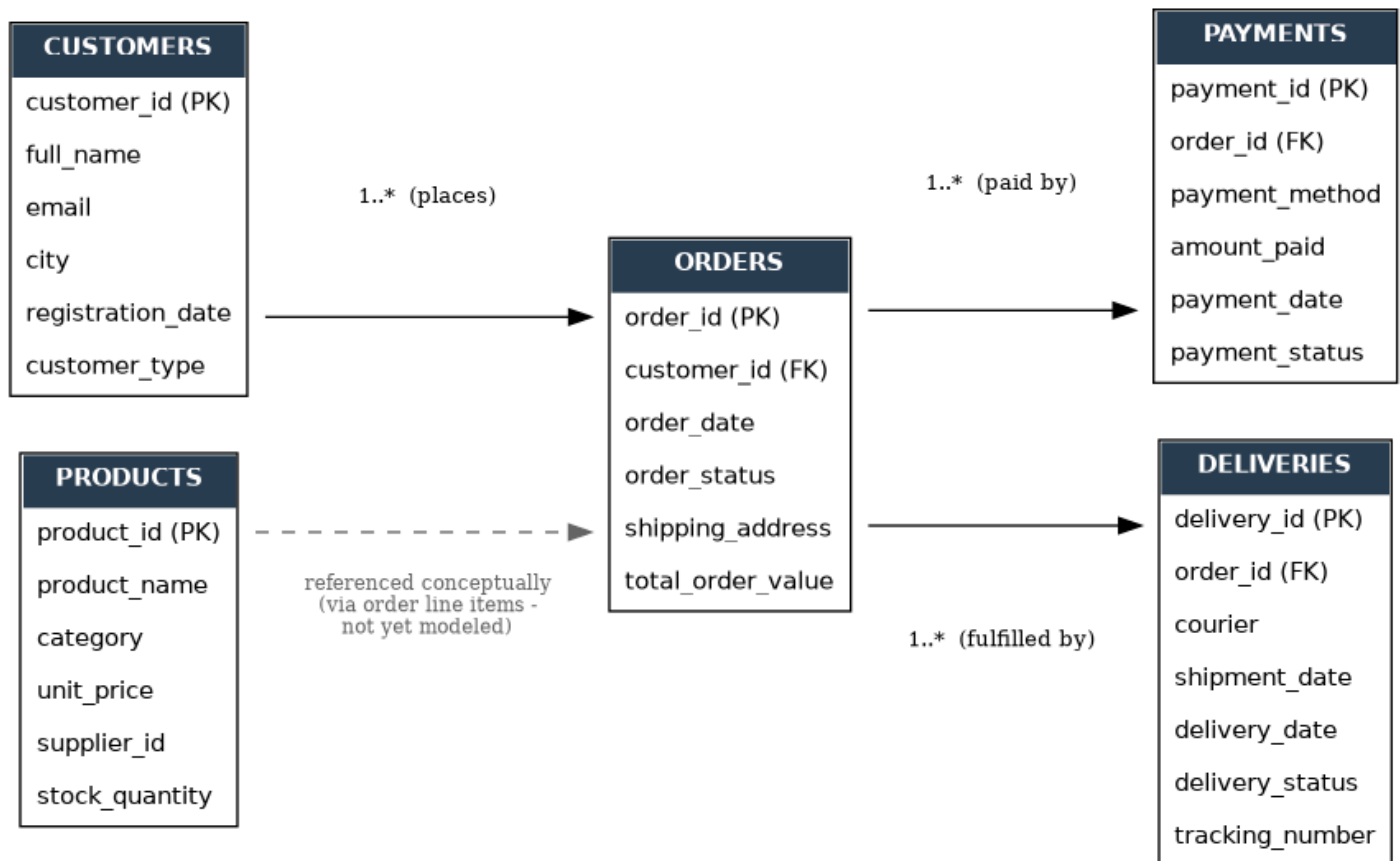
Source System Summary

Source System	Purpose	Main Identifiers	Related System
Customer System	Stores customer profile and data	customer_id	Order Management System
Product Catalog	Stores sellable product Catalog information	product_id	Order Management System
Order Management System	Records what was ordered, by who and current status	order_id	Customers System(customer_id); Payments and Deliveries
Payment Gateway	Stores financial transaction against orders	payment_id	Order Management System
Delivery System	Track shipment and delivery of orders	delivery_id	Order Management System

Task 2 - Identify Entities and Relationships

Identify the major entities in your scenario and draw a simple relationship diagram showing how the systems are likely to connect. This is a relationship draft, not yet a complete formal ERD.

- Label each major entity.
- Show the likely connecting field(s).
- Indicate obvious one-to-many relationships where applicable.
- You may use diagrams.net, PowerPoint, Canva, a handwritten diagram inserted as an image, or another diagramming tool.



Task 3

3.1 Customer System - customers.csv

Field	Expected Type	Key	Nullable?	Description
customer_id	string	PK	No	Unique Identifier for a customer
full_name	string		No	Customer's full name
email	string		No	Customer's email address
city	string		Yes	City associated with the customer
registration_date	date		No	Date the customer account was created
customer_type	string		No	Segment: New, Regular, or Premium

3.2 Product Catalog - product.csv

Field	Expected Type	Key	Nullable?	Description
product_id	string	PK	No	Unique Identifier for a product
product_name	string		No	Name of the product
category	string		No	Product category grouping
unit_price	decimal		No	Selling price per unit
supplier_id	string	FK	Yes	Identifier of the supplying vendor
stock_quantity	integer		No	Units currently available in inventory

3.3 Order Management System - orders.csv

Field	Expected Type	Key	Nullable?	Description
order_id	string	PK	No	Unique Identifier for an order
customer_id	string	FK	No	Customer who placed the order
order_date	date		No	Date the order was placed
order_status	string		No	Confirmed, Pending, Delivered, or Cancelled
shipping_address	string		Yes	Address the order should be delivered to
total_order_value	decimal		No	Total monetary value of the order

3.4 payment gateway - payment.csv

Field	Expected Type	Key	Nullable?	Description
payment_id	string	PK	No	Unique identifier for a payment transaction
order_id	string	FK	No	Order the payment applies to
payment_method	string		No	Credit Card, GCash, Bank Transfer, COD, PayMaya
amount_paid	decimal		No	Amount actually paid
payment_date	date		NO	Date the payment was processed
payment_status	string		No	Successful, Failed, or Pending

3.5 Delivery System — deliveries.csv

Field	Expected Type	Key	Nullable?	Description
delivery_id	string	PK	No	Unique identifier for a delivery record
order_id	string	FK	No	Order the delivery applies to
courier	string		Yes	Courier/carrier handling the shipment
shipment_date	date		Yes	Date the order was shipped (blank if not yet shipped)
delivery_date	date		Yes	Date the order was delivered (blank until delivered)
delivery_status	string		No	Delivered, In Transit, Returned, or Not Shipped
tracking_number	string		No	Courier tracking reference

Task 4

Source	Primary Key(PK)	Possible Foreign Keys	Likely one-to-many relationships
customer	customer_id	None	One customer to many orders
product	product_id	supplier_id	One product to multiple orders
order	order_id	customer_id	One order to many payments, one order to many deliveries
payment	payment_id	order_id	N/A

delivery	delivery_id	order_id	N/A
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Relationships that could create a referential-integrity problem later

- payment.order_id could point to an order that doesn't actually exist in the order table
- delivery.order_id could also point to an order that was never created or somehow got removed
- order.customer_id could point to a customer that isn't in the customer table if the two systems go out of sync

Guide question: What would happen if a foreign key contains a value that does not exist in its parent source?

That row basically becomes an orphan; it can't be matched back to where it's supposed to come from. So if a payment or delivery has an order_id that doesn't exist, it won't show up correctly when you try to join it with the orders table, and it could mess up reports like "total paid" or "orders delivered" without anyone noticing. It's not really a coding problem, it's more of a data problem, so it needs to be caught early instead of just ignored.

Short Reflection

1. Why is it important for a Data Engineer to understand the business scenario before writing the pipeline?

It is important for a Data Engineer to first understand the business scenario and the problem because without knowing the problem, you won't know what the data is actually supposed to show or how it's supposed to be used. You could build a pipeline that runs fine but still gives the wrong answer.

2. Why should raw data be preserved instead of immediately modifying the original files?

Preserving raw data is basically a checkpoint. If a mistake happens later or the engineer needs to investigate something, they can always go back to the original source instead of losing it after making changes.

3. What is the purpose of a staging layer?

The staging layer is where the data goes through a verification process to check if it follows the intended schema before anything else is done to it. It's a way to confirm the data loaded correctly before moving on to actual cleaning or integration.

4. Which field or relationship in your scenario is likely to be the most important when the datasets are eventually integrated?

The most important field is the order_id because it connects the orders to both the payments and the deliveries. Without it, there's no way to check if an order was actually paid and delivered.

5. What potential problem do you expect when you eventually combine your five source systems?
One problem I expect is having payments or deliveries that point to an order_id that doesn't exist.